Chapter 12 exercises

Identify each quantity as an observed statistic, model expectation, or estimator. Check symmetry and positive semidefiniteness for covariance matrices.

CH12-E1

Sample data:

$$3, 5, 5, 9.$$

1. Compute the sample mean.
2. Compute sample variance with denominator $N - 1$.
3. Compute descriptive mean squared deviation with denominator N.
4. Explain the two uses.

CH12-E2

A discrete random variable has mass function

x	0	1	2
$p(x)$	0.2	0.5	0.3

1. Find its expectation.
2. Find $\mathbb{E}[X^2]$ and $\mathrm{Var}(X)$.
3. For $Y = 3X - 2$, find mean and variance.
4. Check $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

CH12-E3

Let $X \sim U(0, L)$, where $L > 0$.

1. Give the density and CDF.
2. For $L = 4$, find $\mathbb{P}(1 < X < 3)$.
3. Derive mean and variance in terms of L, including $\mathrm{Var}(X) = L^2/12$, and interpret scaling.
4. Describe the moments after translating the interval to $[a, a + L]$.

CH12-E4

Joint probabilities:

$$P = \begin{bmatrix} 0.2 & 0.1 \\ 0.3 & 0.4 \end{bmatrix},$$

Rows correspond to $X = 0, 1$, and columns to $Y = 0, 1$.

1. Compute $P(X = 0), P(X = 1), P(Y = 0), P(Y = 1)$.
2. Compute $\mathbb{E}[X], \mathbb{E}[Y], \mathbb{E}[XY]$.
3. Compute covariance and correlation.
4. Decide independence.

CH12-E5

A two-dimensional random vector takes the following three values, each with probability $1/3$.

$$X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad X_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad X_3 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

These subscripts index possible vector values, not three components of the random vector.

1. Find the mean vector.
2. Find the centered possible values.
3. Compute $V = \mathbb{E}[ZZ^T]$ from their outer products.
4. Determine whether it is positive definite or only semidefinite.

CH12-E6

Let

$$X \sim N(\mu, V), \qquad \mu = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$
$$V = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

1. Write the joint density.
2. Give the equation of a density level curve.
3. For $Y = (X_1, 3X_2)^T$, find mean and covariance.
4. Choose an SPD covariance with nonzero off-diagonal entry and write the resulting density.

CH12-E7

Measurements:

$$b_1 = x + u_1, \quad b_2 = x + u_2,$$

Here x is a fixed unknown constant. The error vector $(u_1, u_2)^T$ has mean zero and covariance $V = \mathrm{diag}(\sigma_1^2, \sigma_2^2)$, with $\sigma_1, \sigma_2 > 0$. A diagonal covariance means uncorrelated errors; independence and normality are unnecessary.

1. Write the weighted objective.
2. Derive the parameter estimate.
3. Find its variance.
4. Evaluate the estimate for $\sigma_1 = 1, \sigma_2 = 1.5, b_1 = 10, b_2 = 11$.

CH12-E8

Model $b = Ax + E$, with E zero mean and known SPD covariance V, fixed design A. Let

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad V = \mathrm{diag}(1, 4, 9).$$

1. Write the weighted normal equations.
2. Compute $(A^T V^{-1} A)^{-1}$ symbolically.
3. State the BLUE assumptions and discuss whether normality is required.
4. Describe ways to handle a singular covariance estimate and what must be reported.

CH12-E9

1. Bound the rank of $\widehat{V}_{n-1} = \frac{1}{n-1} X_c X_c^T$, especially for $M \gg n$.
2. Let X have finite second moments, mean μ, and covariance V. Let independent $Z \sim N(0, W)$. Prove the sum has mean μ and covariance $V + W$. State when the sum is also Gaussian.
3. Interpret correlation and covariance rank for two perfectly linearly related variables, including constant-variable cases.
4. Explain why covariance does not identify a causal effect, using a common-cause example.

Lin Cing https://github.com/xinhi6716/computer-science-self-study

CH12-E10

Combine covariance spectral geometry, whitening, and WLS.

1. Let

$$V = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

Verify SPD, find eigenvalues and orthonormal eigenvectors, and state the whitening scale in each direction. 2. Let $b = Ax + e$, where

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad V_e = \mathrm{diag}(1, 4, 9), \qquad b = (2, 5, 4)^T.$$

Form and solve the weighted normal equations, and find the weighted residual and $J(\hat{x})$. 3. For the same A, b, incorrectly setting $V_e = I$, find the OLS estimate and its difference from WLS. 4. Give sufficient assumptions for BLUE. Which steps require normality?

CH12-E11

1. Let X be uniform on $-1, 0, 1$, $Y = X^2$. Find both means, variances, and covariance. Test independence using conditional probability.
2. For a sum S_N of N independent fair 0/1 variables, derive the values, spacing, and probabilities of standardized Z_N. Give the histogram heights suitable for comparison with normal density.
3. Independently draw $X_i \sim U(-1, 1)$, and estimate $1/3$ by $N^{-1} \sum_i X_i^2$. Find its mean and variance, and a sufficient N for Chebyshev to bound the probability of absolute error at least 0.01 by 0.05.
4. With IID observations and finite positive variance alone, does CLT guarantee any prescribed accuracy at $N = 30$? Distinguish a finite-sample bound from asymptotic convergence.

CH12-E12

Estimate one fixed unknown constant x, without state evolution or process noise. The old estimate is unbiased with error variance $W_0 = 1$, and its observed value is $\widehat{x}_0 = 10$. New observations satisfy $b_i = x + \eta_i$, $\mathbb{E}[\eta_i] = 0$, $R_1 = 9, R_2 = 1$, with observed values $b_1 = 13, b_2 = 9$. Assume η_1, η_2 are mutually uncorrelated and each is uncorrelated with the old estimation error $\widehat{x}_0 - x$. Both measurement matrices are $H = 1$.

Parts 1 and 3 also require the vector derivation. Treat the old estimate as an unbiased linear WLS summary of the same fixed state, with $W_0 \succ 0$. The new model has $H \in \mathbb{R}^{m \times n}$, $R = \mathrm{Cov}(\eta) \succ 0$, and new errors uncorrelated with the old estimation error. Normality is unnecessary.

1. Derive $\widehat{x}_1 = \widehat{x}_0 + K(b_1 - H\widehat{x}_0)$ from the combined normal equations, and prove $K = W_0 H^T (R + H W_0 H^T)^{-1}$.
2. Compute the gains, estimates, and variances of both sequential updates, and compare with the batch solution.
3. Prove $W_1 \preceq W_0$. Why must a complete copy of an earlier observation not be counted again using the uncorrelated-error update?

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 12.

Original educational synthesis by Lin Cing, licensed under CC BY 4.0.
https://github.com/xinhi6716/computer-science-self-study